

One Type of Non-linear 2nd Order ODE Form:

$$x'' = kx^n \quad (n \neq -1),$$

where $x(t)$ is a function of t

Multiply both sides by x' :

$$x'x'' = kx^n x'$$

Note that $\frac{d}{dt} \left(\frac{1}{2} (x')^2 \right) = \left(\frac{1}{2} * 2 * x' \right) * x'' = x'x''$, $\frac{d}{dt} (x^{n+1}) = (n+1)x^n x'$

$$x'x'' = kx^n x'$$

$$\frac{d}{dt} \left(\frac{1}{2} (x')^2 \right) = k(n+1)^{-1} \frac{d}{dt} (x^{n+1})$$

Integrate both sides w.r.t t

$$\frac{1}{2} (x')^2 = k(n+1)^{-1} x^{n+1} + C.$$

Initial Condition Form:

$$\text{When } x = a, x' = b$$

Apply initial conditions

$$\frac{1}{2} b^2 = k(n+1)^{-1} a^{n+1} + C,$$

$$C = ?$$